Worksheet — Pulleys and Hanging Masses Simulation

Use with *Simulation - Pulleys and hanging masses.html* · one pulley (Atwood machine) and two or three pulleys with a knot · Newton's laws (Blocks 3–4)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/23-pulleys-hanging-masses/** in a browser (offline: *Simulation - Pulleys and hanging masses.html*). Choose the **number of pulleys** at the top: ① one pulley (Atwood machine) · ② two pulleys and a knot · ③ three pulleys and a knot. Pick an **Example** or set the masses yourself (▼▲, or type a value in the box).

① Press ▶ **Release**; ◂ / ▸ step through the motion. ② ③ **Drag a pulley** or change a mass: the knot swings and settles. Read the angles on the protractor around the knot.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 25 min) · **Part 3** homework (≈ 30 min). g = 9.81 N kg^{-1}. Answers to 3 s.f.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. ① 0.200 kg and 0.220 kg hang over one pulley. While they move, is the tension equal to the lighter weight, between the two weights, or equal to the heavier weight?			
2. ① Two equal masses hang over the pulley. You push one down and let go. Does it slow down, keep a constant speed or speed up?			
3. ② 0.30 kg, 0.40 kg and 0.50 kg (the 0.50 kg hangs from the knot). Is the angle between strings 1 and 2 less than, equal to or more than 90°?			
4. ② Now m_3 = 0.80 kg (m_1 = 0.30 kg, m_2 = 0.40 kg). What happens to the knot?			

5. (Card C2★) Masses of 2.0 kg and 3.0 kg hang over a light, smooth pulley. Calculate (a) the acceleration and (b) the tension. Check: ① → Example *Card C2★*.

Working: FBD of each mass → ΣF = ma for each (or the whole system)

6. (3-4-5 knot) Use question 3: the strings meet at 90°. Calculate the angle between the string of the 0.30 kg mass and the vertical. Check: ② → Example *3-4-5*; read θ_1.

Working: FBD of the knot → $\Sigma F_x = 0$

Part 2 — During the lab

A · Atwood machine: measure g

Pulley on a clamp, two hangers with slotted masses, drop height h = 1.00 m (metre rule), stopwatch or light gate. Keep **$m_1 + m_2 = 0.420$ kg**: move 5 g or 10 g slotted masses from one side to the other. Time the drop three times and take the mean.

7. For 0.200 kg + 0.220 kg, calculate the acceleration a light, smooth pulley would give and the time to fall 1.00 m.

Working

run	m_1 / kg	m_2 / kg	$x = (m_2 - m_1)/(m_1 + m_2)$	t_1 / s	t_2 / s	t_3 / s	mean t / s	$a = 2h/t^2$ / m s^{-2}	sim a / m s^{-2}
1									
2									
3									
4									
5									

8. Plot a (y-axis) against x on graph paper and draw the best-fit line. Gradient = _____ m s^{-2} intercept = _____ m s^{-2}

Compare your gradient with 9.81 m s^{-2}: percentage difference = _____ %. In the simulation: ● Record five runs, Data tab → *a*–*x*.

9. Predict, then test. A real pulley has friction (set *Pulley friction* = 0.02 N and record the five runs again). The a–x line: ☐ is steeper ☐ has the same gradient but moves down ☐ goes through the origin with a smaller gradient. Explain:

Explanation

B · Pulley board: three forces in balance

Two pulleys clamped to a vertical board, three hangers tied to one knot, a protractor behind the knot. Let the knot settle, tap the board gently, then read each string's angle to the vertical (plumb line).

set-up	m_1 / kg	m_2 / kg	m_3 / kg	θ_1 measured	θ_2 measured	θ_1 sim	θ_2 sim	angle 1–2
3-4-5	0.30	0.40	0.50					
equal	0.30	0.30	0.30					
own								
own								

10. For each row, calculate the horizontal components $T_1 \sin\theta_1$ and $T_2 \sin\theta_2$ ($T = mg$) and the vertical total $T_1 \cos\theta_1 + T_2 \cos\theta_2$. Do they show $\Sigma F_x = 0$ and $\Sigma F_y = 0$ within your reading uncertainty (± 1°)?

Working

11. Predict, then test. You move one pulley higher on the board. When the knot settles again, are the angles between the strings different, the same, or all 120°? Drag a pulley in the simulation to check, then explain.

Prediction and explanation

Part 3 — Homework: analyse

12. (Problem set Q27) With $m_1 = 0.190$ kg and $m_2 = 0.210$ kg a student measures $a = 0.48$ m s^{-2}. (a) Determine g from this run. (b) Assume the difference from 9.81 m s^{-2} is caused by friction in the pulley. Calculate the friction force. Check: ① → *Problem set Q27*.

Working

13. (Card C2★★★) Masses of 1.20 kg and 0.80 kg hang over a smooth pulley; the 1.20 kg mass starts 0.60 m above the floor. Calculate its speed when it hits the floor. Check: ① → *Card C2★★★*, step to the end.

Working

14. (FBD guide) Masses of 1.0 kg and 2.0 kg hang over a smooth pulley. Calculate the tension and show that it lies between the two weights.

Working

15. ② $m_1 = 0.30$ kg, $m_2 = 0.40$ kg and $m_3 = 0.60$ kg. Sketch the vector triangle and calculate the angle between strings 1 and 2. Check: ② → *0.30, 0.40, 0.60 kg*.

Working

16. (Problem set Q19) A 15 kg shop sign hangs from two cables, each at 40° to the horizontal. Draw the FBD of the knot and calculate the tension in each cable. Check: ② → *Problem set Q19* (the two 11.67 kg masses play the cables).

Working

17. ③ Three equal masses hang over three pulleys. Explain, using a vector triangle, why the angle between every pair of strings is 120°, wherever the pulleys are (as long as the knot can balance).

Working